

23-2 DHCP Troubleshooting - Answer Key

In this lab you will troubleshoot DHCP.

DHCP Troubleshooting

- 1) The campus LAN has been configured but users in the 10.10.10.0/24 subnet are complaining of connectivity issues.

Your task is to troubleshoot the configuration and ensure that all PCs have connectivity to each other and to the Service Provider router.

R2 has been configured as a DHCP server for PCs in the 10.10.10.0/24 subnet.

The Service Provider router is an external device which you do not have any administrative access to. It has been preconfigured as a DHCP server which will give an IP address to the outside interface on R1. Use show commands to discover the IP address of the Service Provider router.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

First check to see if the PCs have a valid IP address in the 10.10.10.0/24 subnet.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Link-local IPv6 Address.....:
FE80::200:CFF:FEA0:A359
    Autoconfiguration IP Address....: 169.254.163.89
    Subnet Mask.....: 255.255.0.0
    Default Gateway.....: 0.0.0.0
```

The PCs are generating an IP address in the 169.254.0.0 range. This happens when they cannot acquire an IP address via DHCP.

The PCs DHCP server is R2. Check to see if R1 has been configured with an 'ip helper-address' to forward the requests on.

```
R1#show run
Building configuration...

Current configuration : 751 bytes
!
hostname R1
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
interface FastEthernet0/1
ip address 10.10.10.1 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet1/0
ip address 10.10.20.1 255.255.255.0
ip helper-address 10.10.20.10
duplex auto
speed auto
!
truncated
```

R1's FastEthernet1/0 interface has been configured with an ip helper-address of 10.10.20.10. This is the wrong IP address as R2 is available at 10.10.20.2, and it is on the wrong interface because the command must be configured on the interface which receives the requests. Correct the error.

```
R1(config)#int f1/0
R1(config-if)#no ip helper-address 10.10.20.10
R1(config-if)#int f0/1
R1(config-if)#ip helper-address 10.10.20.2
```

Issue the 'ipconfig /renew' command on the PCs to try to acquire an IP address again.

```
C:\>ipconfig /renew
DHCP request failed.
```

The DHCP request is still failing. Check the DHCP configuration on R2.

```
R2#show run
Building configuration...

Current configuration : 758 bytes
!
hostname R2
!
ip dhcp excluded-address 10.10.10.1 10.10.10.10
!
ip dhcp pool Flackbox
network 10.10.10.0 255.255.255.255
default-router 10.10.10.1
dns-server 8.8.8.8
!
truncated
```

The network statement in the DHCP configuration has been misconfigured, it is using 10.10.10.0 with a /32 subnet mask so it cannot give out any valid addresses. Correct the configuration to use the correct mask of /24.

```
R2(config)#ip dhcp pool Flackbox
R2(dhcp-config)#no network 10.10.10.0 255.255.255.255
R2(dhcp-config)#network 10.10.10.0 255.255.255.0
```

Attempt to renew the IP addresses on the PCs again.

```
C:\>ipconfig /renew

IP Address.....: 10.10.10.13
Subnet Mask.....: 255.255.255.0
Default Gateway...: 10.10.10.1
DNS Server.....: 8.8.8.8
```

```
C:\>ipconfig /renew

IP Address.....: 10.10.10.11
Subnet Mask.....: 255.255.255.0
Default Gateway...: 10.10.10.1
DNS Server.....: 8.8.8.8
```

That's working now. Verify the PCs have connectivity to each other.

```
C:\>ping 10.10.10.11

Pinging 10.10.10.11 with 32 bytes of data:

Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

The next thing is to check that they also have connectivity to the Service Provider router, but we don't know its IP address yet. Check the IP address on the outside FastEthernet 0/0 interface on R1.

```
R1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   up          up
FastEthernet0/1 10.10.10.1      YES manual   up          up
FastEthernet1/0 10.10.20.1      YES manual   up          up
FastEthernet1/1 unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down
```

It doesn't have an IP address. Check the configuration.

```
R1#show run
Building configuration...

Current configuration : 751 bytes
!
hostname R1
!
interface FastEthernet0/0
no ip address
duplex auto
speed auto
!
```

It hasn't been configured as a DHCP client. Correct that.

```
R1(config)#int f0/0
R1(config-if)#ip address dhcp
R1(config-if)#
```

```
%DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet0/0 assigned
DHCP address 203.0.113.2, mask 255.255.255.0, hostname R1
```

IP address 203.0.113.2 was assigned. Check the DHCP lease details to discover the IP address of the Service Provider router which is the DHCP server.

```
R1#show dhcp lease
Temp IP addr: 203.0.113.2 for peer on Interface:
FastEthernet0/0
Temp sub net mask: 255.255.255.0
DHCP Lease server: 203.0.113.201 , state: Bound
DHCP Transaction id: C98CBA26
Lease: 86400 secs, Renewal: 43200 secs, Rebind: 75600 secs
Temp default-gateway addr: 203.0.113.1
Next timer fires after: 11:58:27
Retry count: 0 Client-ID:cisco-0001.63C2.9701-Fa0/0
Client-ID hex dump: 636973636F2D303030312E363343322E
93730312D4661302F30
Hostname: R1
```

Verify the PCs have connectivity to the Service Provider router at 203.0.113.201.

```
C:\>ping 203.0.113.201

Pinging 203.0.113.201 with 32 bytes of data:

Request timed out.
Reply from 203.0.113.201: bytes=32 time<1ms TTL=254
Reply from 203.0.113.201: bytes=32 time<1ms TTL=254
Reply from 203.0.113.201: bytes=32 time<1ms TTL=254

Ping statistics for 203.0.113.201:
    Packets: Sent = 4, Received = 3, Lost = 1 (25%
loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Connectivity has now been restored.